WORKED SOLUTION

STEP II 1989, Question 8

Pure • Groups of Transformations

STEP II 1989, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q8

The key idea

Everything rests on one identity: $\Omega^3 = -1$, so Ω satisfies the quadratic in the first line, and that lets us reduce any power of Ω to something linear in Ω. Once we know how R and T shuffle the four points, the counting is about the four points, not the formulae: the group must carry z_0 to every one of the four points, and it already contains three different maps that leave z_0 alone. For the last part, ask what kind of permutation R and T are.

Solution

The identity

Since $\Omega = e^{i\pi/3}$, we have $\Omega^3 = e^{i\pi} = -1$, so

$$0 = \Omega^3 + 1 = (\Omega + 1)(\Omega^2 - \Omega + 1).$$

As $\Omega \neq -1$, the second factor vanishes: $\Omega^2 - \Omega + 1 = 0$, as required.

We will use this in the forms

$$\Omega^2 = \Omega - 1, \qquad \Omega^3 = -1, \qquad \Omega^4 = -\Omega.$$

Note also $2\Omega - 1 = 2\cos\frac{\pi}{3} - 1 + 2i\sin\frac{\pi}{3} = i\sqrt{3} \neq 0$.

R and T permute the four points

For R, multiplying by $\Omega^2 = e^{2i\pi/3}$ is a rotation by 120° about 0:

$$R(0) = 0, \qquad R(1) = \Omega^2, \qquad R(\Omega^2) = \Omega^4 = -\Omega, \qquad R(-\Omega) = -\Omega^3 = 1.$$

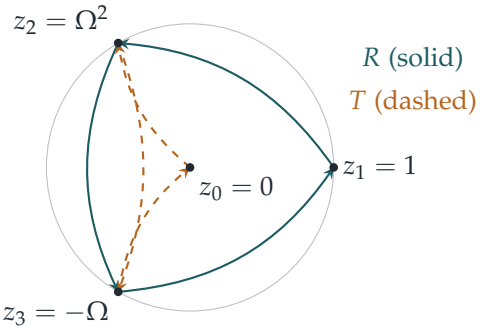
So R fixes z_0 and sends $z_1 \to z_2 \to z_3 \to z_1$.

For T, we substitute each point and simplify with $\Omega^2 = \Omega - 1$:

$$\begin{aligned} T(0) &= \frac{\Omega^2}{1} = z_2, \\ T(1) &= \frac{\Omega + \Omega^2}{2\Omega^2 + 1} = \frac{2\Omega - 1}{2\Omega - 1} = 1 = z_1, \\ T(\Omega^2) &= \frac{\Omega^3 + \Omega^2}{2\Omega^4 + 1} = \frac{\Omega - 2}{1 - 2\Omega} = -\Omega = z_3, \\ T(-\Omega) &= \frac{-\Omega^2 + \Omega^2}{-2\Omega^3 + 1} = \frac{0}{3} = 0 = z_0. \end{aligned}$$

In the third line, $\Omega^3 + \Omega^2 = -1 + (\Omega - 1) = \Omega - 2$, and the last step holds because $-\Omega(1 - 2\Omega) = -\Omega + 2\Omega^2 = -\Omega + 2\Omega - 2 = \Omega - 2$. All the denominators are non-zero (they are 1, $\mathrm{i}\sqrt{3}$, $-\mathrm{i}\sqrt{3}$ and 3).

So T fixes z_1 and sends $z_0 \to z_2 \to z_3 \to z_0$. Both R and T permute the four points; each is a 3-cycle.



The group has order at least 12

Let G be the smallest group of transformations containing R and T: everything we can build from R, T and their inverses by composition. Every element of G maps the set $\{z_0, z_1, z_2, z_3\}$ onto itself, because R and T do (and so do their inverses, being inverses of bijections of this finite set).

Three maps fixing z_0. The set $H = \{I, R, R^2\}$ lies in G, every element fixes z_0, and the three are different transformations (rotations by 0°, 120°, 240°).

Maps sending z_0 everywhere. Using the cycles found above,

$$g_0 = I, \qquad g_1 = R^2T, \qquad g_2 = T, \qquad g_3 = RT$$

satisfy $g_k(z_0) = z_k$: for example $RT(z_0) = R(z_2) = z_3$ and $R^2T(z_0) = R^2(z_2) = R(z_3) = z_1$.

Now consider the twelve products $g_k h$ with $k \in \{0, 1, 2, 3\}$ and $h \in H$. Since $h(z_0) = z_0$, we get $g_k h(z_0) = z_k$.

Ask yourself Could two of these twelve products be the same transformation?

No. If $g_k h = g_l h'$, apply both sides to z_0: we get $z_k = z_l$, so $k = l$. Then composing on the left with g_k^{-1} gives $h = h'$. So the twelve products are distinct elements of G, and $|G| \geq 12$. (In the language of group theory: the orbit of z_0 has size 4 and its stabiliser has size at least 3, so by the orbit–stabiliser theorem $|G| \geq 4 \times 3$.)

The final part

Yes: exactly half of the 24 permutations of the four points can never be produced.

Ask yourself What do R and T have in common as permutations?

Each is a 3-cycle, and a 3-cycle is an *even* permutation: for instance $(z_1\, z_2\, z_3) = (z_1\, z_3)(z_1\, z_2)$, a product of two transpositions. Their inverses are also 3-cycles. A composition of even permutations is even, so every permutation of the points produced by R and T is even. There are only 12 even permutations of four objects, so the other 12, the odd ones, are never produced. For example, swapping z_0 and z_1 while fixing z_2 and z_3 is a single transposition, hence odd, and cannot be obtained.

In fact, combined with the previous part this pins G down exactly. R and T are Möbius maps (T has $ad - bc = \Omega - 2\Omega^4 = 3\Omega \neq 0$), and a Möbius map fixing three points is the identity, so each element of G is determined by how it permutes the four points. Hence $|G| \leq 12$, and with the previous part, $|G| = 12$: G acts on the points as the full group of even permutations, A_4.

Checks

- Numerically, $|\Omega^2 - \Omega + 1| \approx 2 \times 10^{-16}$, and evaluating R and T at the four points gives exactly the two 3-cycles above.
- $R^3 = I$ obviously; and $T^3 = I$ too, as a 3-cycle should (checked at a random point, to 10^{-16}).
- Generating all compositions of R and T by computer gives exactly 12 distinct maps, all inducing even permutations, with z_0 moved to all four points and 3 maps fixing it.

Common slips

- Arguing "G contains elements of order 3, so $|G| \geq 12$". Elements of order 3 alone only give 3 dividing $|G|$; it is the transitivity on four points that supplies the factor 4.
- Leaving powers of Ω unsimplified, so that $T(\Omega^2) = -\Omega$ looks false. Reduce everything to the form $a + b\Omega$ first.

Beyond STEP

The count $|G| = (\text{orbit size}) \times (\text{stabiliser size})$ is the orbit–stabiliser theorem, and the even permutations form the alternating group A_4, also the rotation group of a regular tetrahedron. There is a lovely reason this happens here: Möbius maps preserve the cross-ratio of four points, and for these four points the cross-ratio is $e^{-i\pi/3}$, the special "equianharmonic" value for which exactly the even permutations preserve it. For groups and group actions, see Allenby, *Rings, Fields and Groups*; for Möbius maps and the Riemann sphere, Needham, *Visual Complex Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.